Motor Control Project Documentation

Before we understand anything about the hardware or firmware we are going to use , we need to understand some basic concepts about 3-phased motors.
Motors can be wound in 2 different manners , Y-winding and Delta-winding.

Now , let us try and understand some mathematics behind these wirings and the impact of input voltage in both the cases.

Y-winding

When the U winding is connected to 12V by turning on the switch U and then V winding is connected to Ground by turning on the switch V. A voltage divider kind of circuit gets made in between , with the midpoint connected to Neutral of the Motor

so $V_n = V_{bus} * Z_v(Z_u + Z_v)$

for the sake of simplicity , $Z_u = Z_v$

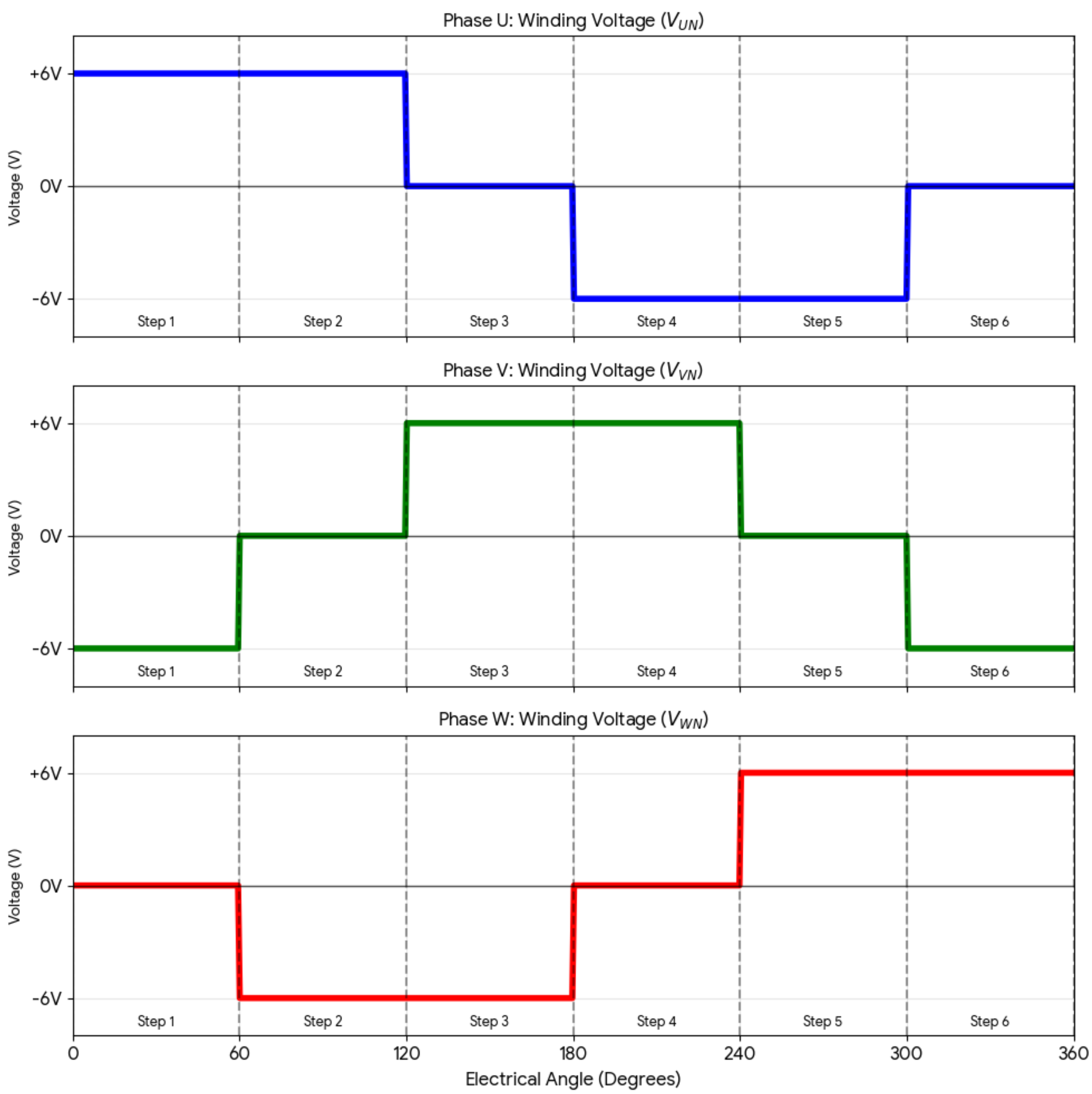
hence $V_n = V_{bus} * Z_v/2Z_v$

$V_n = V_{bus}/2$, hence $V_n = 6V$ given bus voltage is 12V

Now the switches are switched in the following manner

Step	Switch High Side ON	Switch Low Side On(Gnd)	Switch floating	Voltage of Each Phase wrt to Gnd	Voltage of each phase wrt to neutral
U-V (0-60deg)	U	V	W	Vu = 12V, Vv = 0V , Vw = 6V	Vun = 6V Vvn = -6V Vwn = 0V
U-W (60-120deg)	U	W	V	Vu = 12V, Vw = 0V , Vv = 6V	Vun = 6V Vwn = -6V Vvn = 0
V-W (120-180 deg)	V	W	U	Vv = 12V, Vw = 0V , Vu = 6V	Vvn = 6V Vwn = -6V Vun = 0V
V-U (180-240degdeg)	V	U	W	Vv = 12V, Vu = 0V , Vw = 6V	Vvn = 6V Vun = -6V Vwn = 0V
W-U (240-300deg)	W	U	V	Vw = 12V, Vu = 0V , Vv = 6V	Vwn = 6V Vun = -6V Vvn = 0V
W-V (300-360deg)	W	V	U	Vw = 12V, Vv = 0V , Vu = 6V	Vwn = 6V Vvn = -6V Vun = 0V

Now , if you plot the graph of phase to neutral voltages , it will look something like the below image



okay , so now the next big thing is to know the step we are currently in
imagine your system boots up , your motor is in some step 4 , and if you turn on the wrong igbt , the system will be unpredictable.
so the very first step is to measure the present position of the motor by using either hall effect sensors or back emf
but what if there are no sensors??

in that case we need to first drive the motor to a known position or we can make use of inductance measuring by sending very high frequency current signals???

next step is to generate a 20Khz frequency PWM on all the top channels and duty cycle willl change based on speed , initilay 0 %

Function Group	Function	Pin	Logic / Peripheral	Possible?
PWM (Drive)	Phase U PWM	PA8	TIM1_CH1 (Advanced Timer)	YES
PWM (Drive)	Phase V PWM	PA9	TIM1_CH2 (Advanced Timer)	YES
PWM (Drive)	Phase W PWM	PA10	TIM1_CH3 (Advanced Timer)	YES
Enable (Gate)	EN Phase U	PC7	Standard GPIO (Output)	YES
Enable (Gate)	EN Phase V	PB6	Standard GPIO (Output)	YES
Enable (Gate)	EN Phase W	PA11	Standard GPIO (Output)	YES
Rotor Pos (Hall)	Hall Sensor 1 (H1)	PA15	TIM2_CH1 (Input Capture)	YES .. only needed when motor has HALL sensors
Rotor Pos (Hall)	Hall Sensor 2 (H2)	PB3	TIM2_CH2 (Input Capture)	YES only needed when motor has HALL sensors
Rotor Pos (Hall)	Hall Sensor 3 (H3)	PB10	TIM2_CH3 (Input Capture)	YES only needed when motor has HALL sensors
BEMF (Voltage)	BEMF Phase U	PA1	ADC1_IN6	YES
BEMF (Voltage)	BEMF Phase V	PA0	ADC1_IN5	YES
BEMF (Voltage)	BEMF Phase W	PA2	ADC1_IN7	YES
Current (Amps)	CURR_FDBK1	PA3	ADC1_IN8	YES
Current (Amps)	CURR_FDBK2	PC2	ADC1_IN3 (Direct)	YES
Current (Amps)	CURR_FDBK3	PC3	ADC1_IN4 (Direct)	YES

Function Group	Function	Pin	Logic / Peripheral	Possible?
Current (Amps)	CURR_REF	PB11	TIM2_CH4 (PWM to DAC Reference)	YES
Safety/Power	Enable / Fault	PC4	GPIO (Input / Interrupt)	YES
Safety/Power	Vbus Monitor	PC0	ADC1_IN1	YES
Safety/Power	NTC (Temp)	PC1	ADC1_IN2	YES
Safety/Power	Power Keep-Alive	PA4	GPIO (Must be HIGH for Nucleo SMPS)	MAYBE NOT NEEDED
Telemetry	PC Link (TX)	PA2	UART2_TX	NOT REQUIRED
Telemetry	PC Link (RX)	PA3	UART2_RX	NOT REQUIRED